

Introduction to IF STATEMENT

ISA 212 basisdata 2

week 4

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Introduction

In real life, we make decisions based on the conditions. For example, look at the following conditions.

If I get a performance bonus this year, I will go for international vacation or else I'll take a domestic vacation

If the weather becomes good, I will plan to go on a bike trip or else I won't

In these examples, we decide as per the conditions. For example, if I get a bonus then only I will go for an international vacation else I will go for domestic vacations. We need to incorporate these conditions-based decisions in programming logic as well. SQL Server provides the capability to execute real-time programming logic using SQL IF Statement.

Syntax

In the following SQL IF Statement, it evaluates the expression, and if the condition is true, then it executes the statement mentioned in IF block otherwise statements within ELSE clause is executed .

```
IF (Expression )
```

```
BEGIN
```

```
-- If the condition is TRUE then execute the following statement
```

```
True Statements;
```

```
END
```

```
ELSE
```

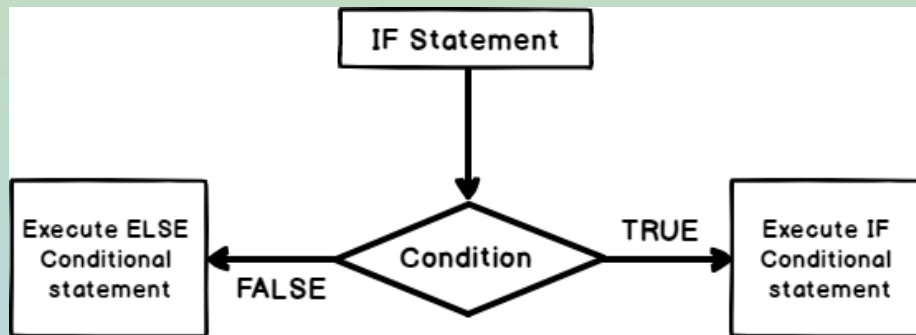
```
BEGIN
```

```
-- If the condition is False then execute the following  
statement
```

```
False Statements
```

```
END
```

FLOW CHART



The condition in SQL IF Statement should return a Boolean value to evaluate
We can specify a Select statement as well in a Boolean expression, but it should enclose in parentheses

We can use BEGIN and END in the IF Statement to identify a statement block

The ELSE condition is optional to use

Example 2: IF Statement with a variable in a Boolean expression

In the following example, we use a variable in the Boolean expression to execute the statement based on the condition. For example, if a student obtained more than 80% marks, he passes an examination else, he is failed.

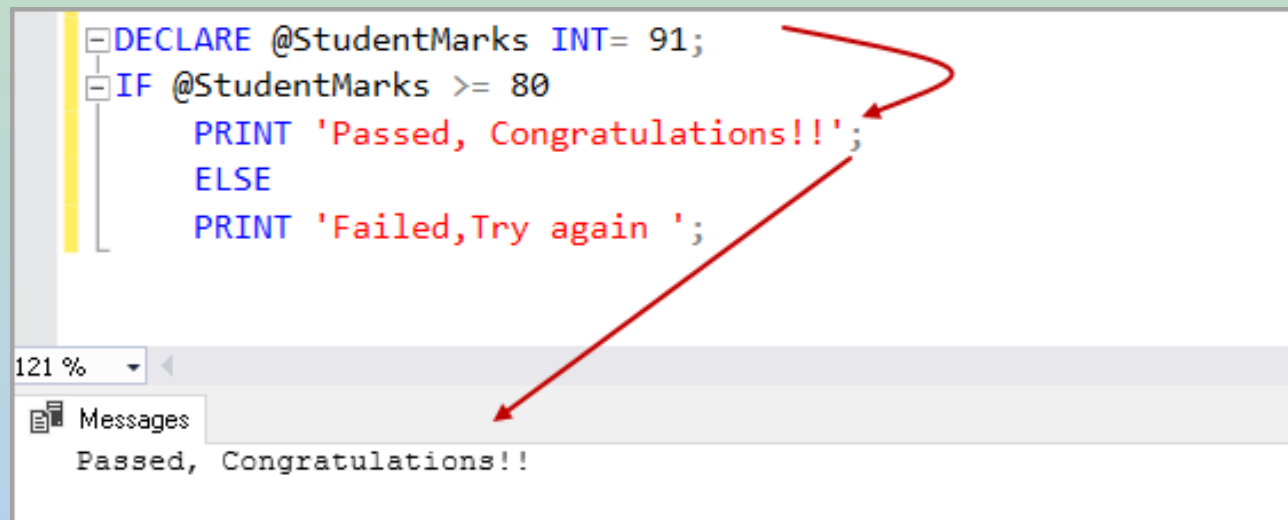
```
DECLARE @StudentMarks INT= 91;  
IF @StudentMarks >= 80  
    PRINT 'Passed, Congratulations!!';  
ELSE  
    PRINT 'Failed, Try again ';
```

Example 2: IF Statement with a variable in a Boolean expression

In the following example, we use a variable in the Boolean expression to execute the statement based on the condition. For example, if a student obtained more than 80% marks, he passes an examination else, he is failed.

```
DECLARE @StudentMarks INT= 91;  
IF @StudentMarks >= 80  
    PRINT 'Passed, Congratulations!!';  
ELSE  
    PRINT 'Failed, Try again ';
```

Example 2: IF Statement with a variable in a Boolean expression



The screenshot shows a SQL query execution window. The query is as follows:

```
DECLARE @StudentMarks INT= 91;  
IF @StudentMarks >= 80  
    PRINT 'Passed, Congratulations!!';  
ELSE  
    PRINT 'Failed, Try again ';
```

Below the query, the output is displayed in a 'Messages' pane. The output is:

```
Passed, Congratulations!!
```

Two red arrows are drawn on the image. One arrow points from the variable `@StudentMarks` in the `IF` statement to the value `91` in the `DECLARE` statement. The other arrow points from the `PRINT 'Passed, Congratulations!!';` line to the output message in the Messages pane.

Example 3: Multiple IF Statement with a variable in a Boolean expression

We can specify multiple SQL IF Statements and execute the statement accordingly. Look at the following example

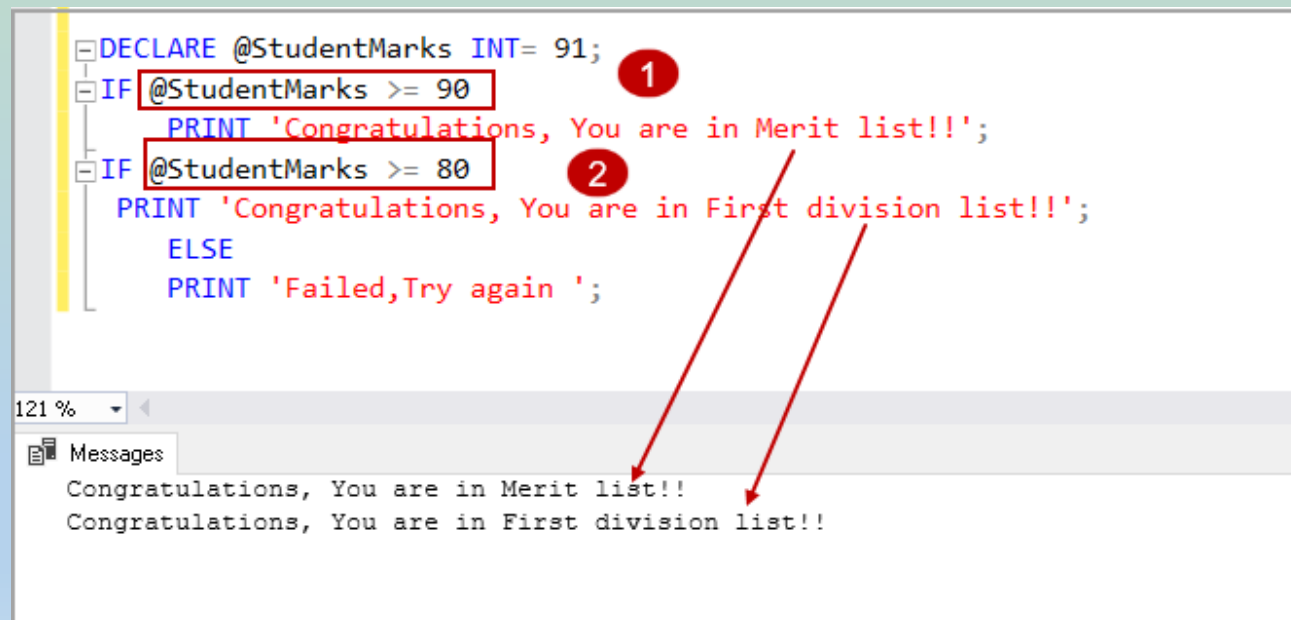
- a. If a student gets more than 90% marks, it should display a message from the first IF statement
- b. If a student gets more than 80% marks, it should display a message from the second IF statement
- c. Otherwise, it should print the message mentioned in ELSE statement

Example 3: Multiple IF Statement with a variable in a Boolean expression

```
DECLARE @StudentMarks INT= 91;
IF @StudentMarks >= 90
    PRINT 'Congratulations, You are in Merit list!!';
IF @StudentMarks >= 80
    PRINT 'Congratulations, You are in First division list!!';
ELSE
    PRINT 'Failed, Try again ';
```

Example 3: Multiple IF Statement with a variable in a Boolean expression

In this example, student marks 91% satisfy the conditions for both SQL IF statements, and it prints a message for both SQL IF statements.



```
DECLARE @StudentMarks INT= 91;
IF @StudentMarks >= 90
    PRINT 'Congratulations, You are in Merit list!!';
IF @StudentMarks >= 80
    PRINT 'Congratulations, You are in First division list!!';
ELSE
    PRINT 'Failed, Try again ';
```

121 %

Messages

Congratulations, You are in Merit list!!

Congratulations, You are in First division list!!

Example 3: Multiple IF Statement with a variable in a Boolean expression

We do not want the condition to satisfy both SQL IF statements. We should define the condition appropriately.

```
DECLARE @StudentMarks INT= 91;
IF @StudentMarks >= 90
    PRINT 'Congratulations, You are in Merit list!!';
IF @StudentMarks >= 80 and @StudentMarks < 90
    PRINT 'Congratulations, You are in First division list!!';
ELSE
    PRINT 'Failed, Try again ';
```

Example 3: Multiple IF Statement with a variable in a Boolean expression

In the output, we can see the following

First, IF statement condition is TRUE. It prints the message inside the IF statement block

Second, IF statement condition is FALSE, it does not print the message inside IF statement block

It executes the ELSE statement and prints the message for it. In this case, we have two SQL IF statements. The second IF statement evaluates to false, therefore, it executes corresponding ELSE statement

Example 3: Multiple IF Statement with a variable in a Boolean expression

```
DECLARE @StudentMarks INT= 91;  
IF @StudentMarks >= 90 1  
    PRINT 'Congratulations, You are in Merit list!!';  
IF @StudentMarks >= 80 and @StudentMarks < 90 2  
    PRINT 'Congratulations, You are in First division list!!';  
ELSE  
    PRINT 'Failed, Try again ';
```

121 %

Messages

Congratulations, You are in Merit list!!
Failed, Try again

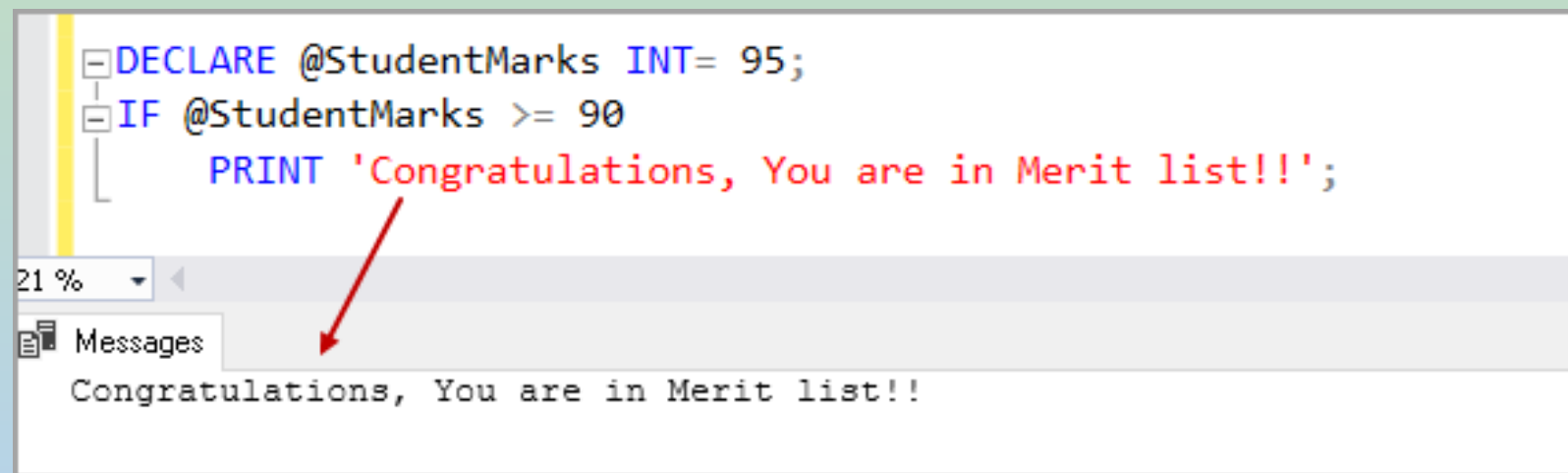
Example 4: IF Statement without ELSE statement

We specified above that Else statement is optional to use. We can use SQL IF statement without ELSE as well.

In the following, the expression evaluates to TRUE; therefore, it prints the message.

```
DECLARE @StudentMarks INT= 95;  
IF @StudentMarks >= 90  
    PRINT 'Congratulations, You are in Merit list!!';
```

Example 4: IF Statement without ELSE statement



```
DECLARE @StudentMarks INT= 95;  
IF @StudentMarks >= 90  
    PRINT 'Congratulations, You are in Merit list!!';
```

21 %

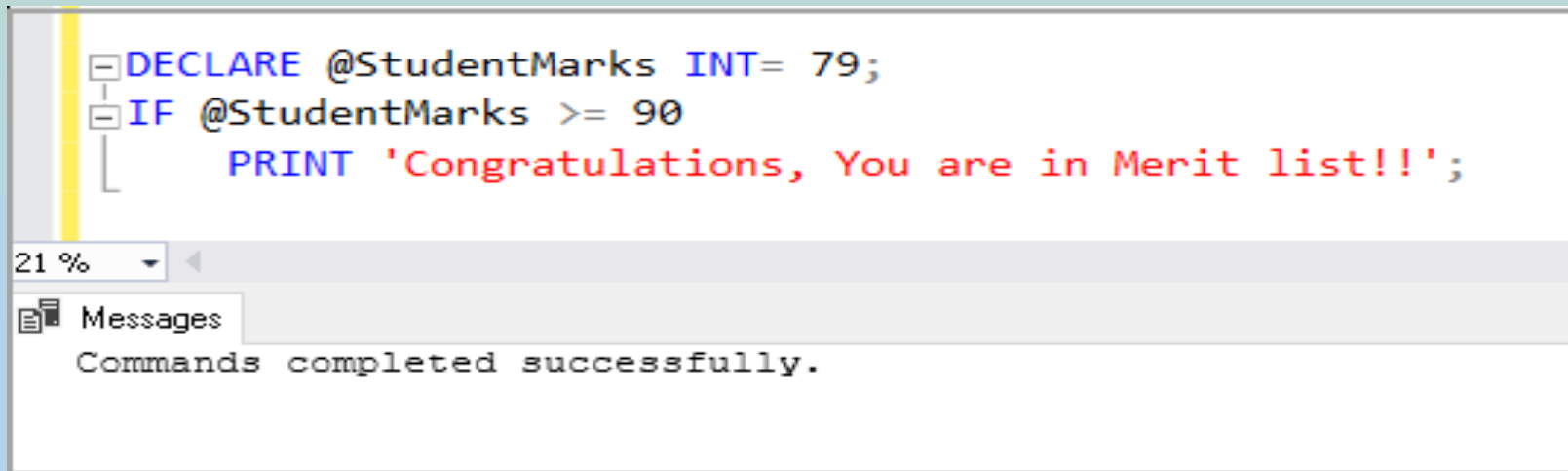
Messages

Congratulations, You are in Merit list!!

A red arrow points from the PRINT statement in the code to the output message in the Messages pane.

Example 4: IF Statement without ELSE statement

if the expression evaluates to FALSE, it does not return any output. We should use ELSE statement so that if an evaluation is not TRUE, we can set default output.



```
DECLARE @StudentMarks INT= 79;  
IF @StudentMarks >= 90  
    PRINT 'Congratulations, You are in Merit list!!';
```

21 %

Messages
Commands completed successfully.

Example 6: IF with BEGIN and END block

We can use BEGIN and END statement block in a SQL IF statement. Once a condition is satisfied, it executes the code inside the corresponding BEGIN and End block.

```
DECLARE @StudentMarks INT= 70;
IF @StudentMarks >= 90
    BEGIN
        PRINT 'Congratulations, You are in Merit list!!';
    END;
ELSE
    BEGIN
        PRINT 'Failed, Try again ';
    END;
```

Example 6: IF with BEGIN and END block

```
DECLARE @StudentMarks INT= 70;  
IF @StudentMarks >= 90  
    BEGIN 1  
        PRINT 'Congratulations, You are in Merit list!!!';  
    END;  
ELSE  
    BEGIN 2  
        PRINT 'Failed, Try again ';  
    END;
```

1 %

Messages

Failed, Try again

Example 6: IF with BEGIN and END block

We can specify multiple statements as well with SQL IF statement and BEGIN END blocks. In the following query, we want to print a message from two print statements once a condition is satisfied.

Note: We should have an END statement with corresponding BEGIN block.

Example 6: IF with BEGIN and END block

```
DECLARE @StudentMarks INT= 70;
IF @StudentMarks >= 90
    BEGIN
        PRINT 'Congratulations, You are in Merit list!!';
        Print 'Second statement.'
    END;
ELSE
    BEGIN
        PRINT 'Failed, Try again ';
        Print 'Second ELSE statement'
    END;
```

Example 6: IF with BEGIN and END block

```
DECLARE @StudentMarks INT= 70;
IF @StudentMarks >= 90
    BEGIN
        PRINT 'Congratulations, You are in Merit list!!!';
        Print 'Second statement'
    END;
ELSE
    BEGIN
        PRINT 'Failed, Try again ';
        Print 'Second ELSE statement'
    END;
```

21 %

Messages

Failed, Try again
Second ELSE statement

STORE PROCEDURE WITH IF STATEMENT